

Title: Government Safety Nets as Catalysts for Climate Adaptation and Women's Empowerment in Agriculture: Insights from Mbire District, Zimbabwe

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Abstract

Climate change presents critical challenges worldwide, particularly for agricultural communities heavily dependent on climate-sensitive livelihoods. Mbire District in Zimbabwe exemplifies such community, where women-who play central roles in agriculture but often lack access to resources and decision-making power-face an outsized impact from climate change. This study examines the effectiveness of government safety nets in supporting climate change adaptation and promoting women's empowerment within agricultural settings, focusing on Mbire District as a case study. A mixed-methods approach was employed, integrating quantitative analyses of government safety net programmes with qualitative insights into their influence on women's empowerment and adaptive capacity. Data were collected through surveys, interviews, and focus group discussions involving women farmers, government officials, and community leaders. The study further assessed the extent to which these programmes address the specific needs of women in agriculture and bolster their resilience to climate-related challenges. Guided by a theoretical framework grounded in feminist perspectives and intersectionality, the research explores the unique vulnerabilities and strengths of women engaged in agricultural livelihoods under changing climatic conditions. By analysing government safety nets through this lens, the study identifies critical opportunities to enhance women's agency and resilience. The findings offer significant contributions to both academic scholarship and policy development, centering the voices and experiences of women farmers in Mbire District as they navigate the intersecting challenges of climate change and gender inequality.

Key Words: Climate change, safety nets, women's empowerment, food security, gender equality, resilience.

Practical Implications

Government safety nets hold significant potential as transformative tools for climate adaptation and the empowerment of women in agriculture, especially in vulnerable regions like Mbire District, Zimbabwe. This chapter provides actionable insights for policymakers and practitioners, focusing on bridging the gap between policy frameworks and practical implementation to ensure these programs deliver tangible benefits to women farmers.

A key practical implication is the integration of climate-resilient agricultural practices into safety net programmes. For instance, as highlighted by Smith et al. (2022), governments could offer subsidized drought-resistant seeds or provide training in conservation agriculture. Incorporating agricultural support into social protection systems has been shown to significantly improve productivity and resilience among smallholder farmers. In Mbire District, where women bear a disproportionate burden from climate shocks, such targeted interventions can not only provide immediate relief but also foster long-term empowerment. By equipping women with essential resources and knowledge, these programmes enable them to enhance agricultural yields and assume vital roles as agents of change within their communities.

The design of inclusive financial mechanisms within safety nets is critically important for advancing both climate adaptation and women's empowerment in agriculture. Johnson and Mwangi (2021), emphasize that cash transfers or microcredit schemes tailored specifically for women farmers can enable them to invest in adaptive technologies, such as solar-powered irrigation systems and improved post-harvest storage facilities. Access to financial resources serves as a cornerstone of women's economic empowerment, allowing them to make strategic decisions and reduce dependency. In resource-constrained areas like Mbire District, such mechanisms not only enhance resilience to climate variability but also promote economic independence among women farmers.

Safety nets can also serve as platforms for fostering collaborative, community-based adaptation strategies. Brown et al. (2020), highlight the potential of women-led farmer cooperatives to enhance knowledge-sharing and collective action. Research underscores the significance of collective management of shared resources, as community-based approaches often yield more sustainable solutions to local challenges. In Mbire District, these cooperatives could play a pivotal role in disseminating best practices for water management, pest control, and soil conservation. This ensures that adaptation strategies are context-specific and aligned regional climate realities.

To maximize the effectiveness of safety net programmes, integrating robust monitoring and evaluation (M&E) systems is crucial. As Patel and Lee (2023) emphasize, gender-sensitive M&E frameworks are essential for identifying and addressing the unique challenges women face in social protection programmes. Such systems enable continuous tracking of progress and facilitate real-time adaptation of strategies. For example, an M&E framework in Mbire District could highlight whether women are receiving equitable access to resources or if cultural norms are

limiting their participation. This feedback loop allows policymakers to refine interventions, ensuring they effectively address the needs of women farmers.

Engaging women beneficiaries through participatory approaches is another critical component. Cornwall et al. (2022) advocate for methods such as focus group discussions and community scorecards, which empower women to share their experiences, challenges, and recommendations. These participatory approaches not only enhance programme effectiveness but also foster a sense of ownership, they provide valuable insights into barriers such as cultural norms, resource constraints, and exclusionary practices that may undermine the success of safety nets.

Evidence from various regions highlights the transformative potential of well-designed safety net programmes. For instance, research by Ahmed et al. (2022) illustrates how social protection initiatives in Sub-Saharan Africa have not only improved food security but also empowered women to adopt climate-resilient agricultural practices. Similarly, Ethiopia's Productive Safety Net Programme (PSNP) demonstrates the importance of integrating gender-sensitive measures to enhance household resilience and foster women's empowerment (World Food Programme, 2021). These case studies provide valuable insights for designing and implementing effective safety net programmes in Mbire District.

In the specific context of Mbire District, Zimbabwe, safety net programmes should emphasize capacity-building initiatives that directly address the unique needs of women. As Ndlovu and Chikukwa (2023) argue, providing training in climate-smart agriculture, facilitating access to microfinance, and establishing robust market linkages can significantly enhance women's adaptive capacity. Overcoming structural barriers, such as unequal access to resources and limited decision-making power, is crucial for achieving sustainable development outcomes. For example, equipping women farmers with skills in crop diversification and ensuring access to markets for surplus produce can enable them to better withstand climate shocks while simultaneously improving household income and livelihoods.

Aligning government safety nets with climate adaptation goals and gender empowerment strategies offers a powerful pathway for fostering resilience and equity. Empowering women in agriculture yields significant multiplier effects on community development, given their pivotal role in household food security and resource management (FAO & UN Women, 2023). Policymakers can leverage these insights to design interventions that translate high-level policies into tangible benefits. By addressing the unique vulnerabilities and capacities of women, these initiatives can enhance livelihoods, strengthen community resilience, and advance sustainable development not only in Mbire District but also in similar contexts across the region.

In conclusion, aligning government safety nets with gender-sensitive and climate-resilient strategies provides an opportunity to address the dual challenges of climate adaptation and women's empowerment in agriculture. Policymakers and practitioners must prioritize participatory approaches, capacity building, and robust M&E systems to ensure equitable and sustainable outcomes. By leveraging these strategies, safety net programmes can transform

women into change agents, fostering resilience and resilience in Mbire District and beyond. A call to action is now essential -to translate these recommendations into tangible investments and ensure that no women farmer is left behind in the fight against climate change.

1.Introduction

Climate change presents a profound threat to agricultural productivity, food security, and the livelihoods of smallholder farmers, especially in developing countries such as Zimbabwe. With increased frequency and intensity of droughts, erratic rainfall patterns, and extreme temperatures, these climatic shifts disrupt traditional farming cycles and exacerbate food insecurity, affecting vulnerable communities disproportionately (FAO, 2018; IPCC, 2022). Recognizing these challenges, the Zimbabwean government has introduced various safety net interventions aimed at strengthening climate change adaptation strategies and promoting women's empowerment in agriculture, recognizing both as essential components of sustainability rural development.

The study focuses on Mbire District, a semi-arid area in northern Zimbabwe, to evaluate the effectiveness of these government-led interventions. Mbire District is particularly susceptible to climate risks, making it an instructive case for assessing how targeted safety nets contribute to climate resilience and the empowerment of women in agriculture. Adaptation strategies in agriculture include the adoption of climate-smart agricultural practices and technologies that enable farmers to manage climate-related risks more effectively (UNDP, 2021; FAO, 2019). These practices include drought-resistant crop varieties, improved irrigation techniques and agroforestry, which together bolster the community's resilience to adverse climatic conditions.

Women in Zimbabwean agriculture face unique, gender-specific challenges that hinder their ability to adapt to climate change fully and access critical agricultural resources and services (Agarwal, 2018; Mutopo, 2021). Structural constraints—such as limited land rights, restricted access to credit, and cultural biases—frequently limit their participation and decision-making power within agricultural settings (World Bank, 2020). As research indicates, empowering women in agriculture is not only crucial for building climate resilience but also for achieving broader goals of food security and sustainable development (Arora-Jonsson, 2019). Thus, understanding the role of government interventions in facilitating women's empowerment requires examining how these programmes address existing gender disparities and promote equitable access to resources.

This research aims to conduct a comprehensive analysis of key factors that influence the success of government safety nets in Mbire District. These factors include the implementation of climate-smart agricultural practices, women farmers' access to resources and support services, the policy frameworks promoting gender equity in agriculture, and the subsequent impact on both agricultural productivity and community resilience. Furthermore, this study integrates perspectives from local communities, government agencies, non-governmental organizations, and other stakeholders involved in climate change adaptation and gender empowerment initiatives (UN Women, 2022; Chigavazira and Mapira, 2020). By synthesizing these diverse perspectives, the study offers a holistic understanding of the opportunities and challenges surrounding government safety nets as catalysts for climate adaptation and women's empowerment in Mbire District.

The insights gained from this research have implications for policymakers and practitioners working to design inclusive climate adaptation frameworks, particularly in regions facing similar

environmental and socio-economic challenges. This research contributes to the growing literature on the intersection of climate resilience, agricultural development, and gender equality by providing evidence on the role of government safety nets in empowering rural women and enhancing community adaptation to climate change in Zimbabwe.

1.1 Background

Climate change encompasses significant, long-term alterations in weather patterns, often evidenced by increased frequency and intensity of droughts, floods, and other extreme weather events. These changes are primarily driven by global warming—an increase in temperatures resulting from greenhouse gas emissions (Shumba, 2012). The impacts of climate change disproportionately affect developing countries, with regions like Africa, Asia, and Latin America anticipated to endure the most severe consequences. According to the Intergovernmental Panel on Climate Change (IPCC, 2007), Africa is especially vulnerable due to its geographical positioning as the hottest continent, with projections indicating that it will warm at a rate 1.5 times faster than the global average. As part of the African continent, Zimbabwe is experiencing similar climate stresses. Since 1900, Zimbabwe's surface temperatures have increased by 0.4 degrees Celsius, underscoring the ongoing environmental shift (Brazier, 2015; Ndlovu, Prinsloo and Le Roux, 2020).

While everyone is susceptible to the effects of climate change, levels of vulnerability vary based on gender, age, and socioeconomic conditions. Rural communities in Zimbabwe, such as those in Mbire District, are more susceptible to climate change compared to urban populations, largely due to higher poverty rates and dependence on agriculture. About 76% of Zimbabwe's rural population lives below the poverty datum line, making them particularly vulnerable to environmental shifts that disrupt agriculture, the primary livelihood in these areas (Brazier, 2015). Women, who make up a significant proportion of the world's impoverished population, bear the brunt of climate change's impacts. According to Brazier (2015), women are especially vulnerable because they rely heavily on natural resources, which are increasingly threatened by climate variability and degradation.

Socially constructed gender roles further intensify women's vulnerability to climate change. Men are often engaged in off-farm work and capitalist enterprises, while women typically handle unpaid tasks like securing food, water, and firewood—activities that expose them more directly to environmental shifts. This gendered division of labor is articulated by Marxist Feminist theory, which highlights how women's roles in unpaid labor contribute to their economic marginalization. As a result, women are disproportionately reliant on natural resources and are thus more exposed to the adverse impacts of climate change.

The concept of eco-feminism further explores the relationship between women and nature, positing that women's close connection to the environment makes them especially vulnerable to environmental degradation. Feminist theory suggests that women suffer disproportionately from ecological issues such as deforestation, which forces them to travel long distances in search of

resources like firewood. Given their dependence on agriculture, environmental degradation also jeopardizes food and nutritional security, thereby amplifying the challenge of securing food (Shiva, 1989; Mies and Shiva, 2014).

Moreover, women's exclusion from decision-making processes exacerbates their vulnerability to climate change. Rural women, particularly, face barriers in social, political, and economic spheres, limiting their capacity to adapt to climate impacts. Brazier (2015) emphasizes that male-dominated structures often marginalize women, restricting their ability to participate in decisions that affect their lives and communities. As a result, women frequently lack the power and confidence needed to influence adaptive strategies, increasing their susceptibility to climate-induced hardships.

Given these intersecting issues, this study seeks to examine the role of government safety nets as catalysts for climate adaptation and women's empowerment in agriculture within Mbire District, Zimbabwe. Understanding the effectiveness of these interventions can provide insights into how government policies might enhance food security and strengthen women's resilience to climate change in rural communities.

1.2 Study Area Description: Mbire District, Zimbabwe

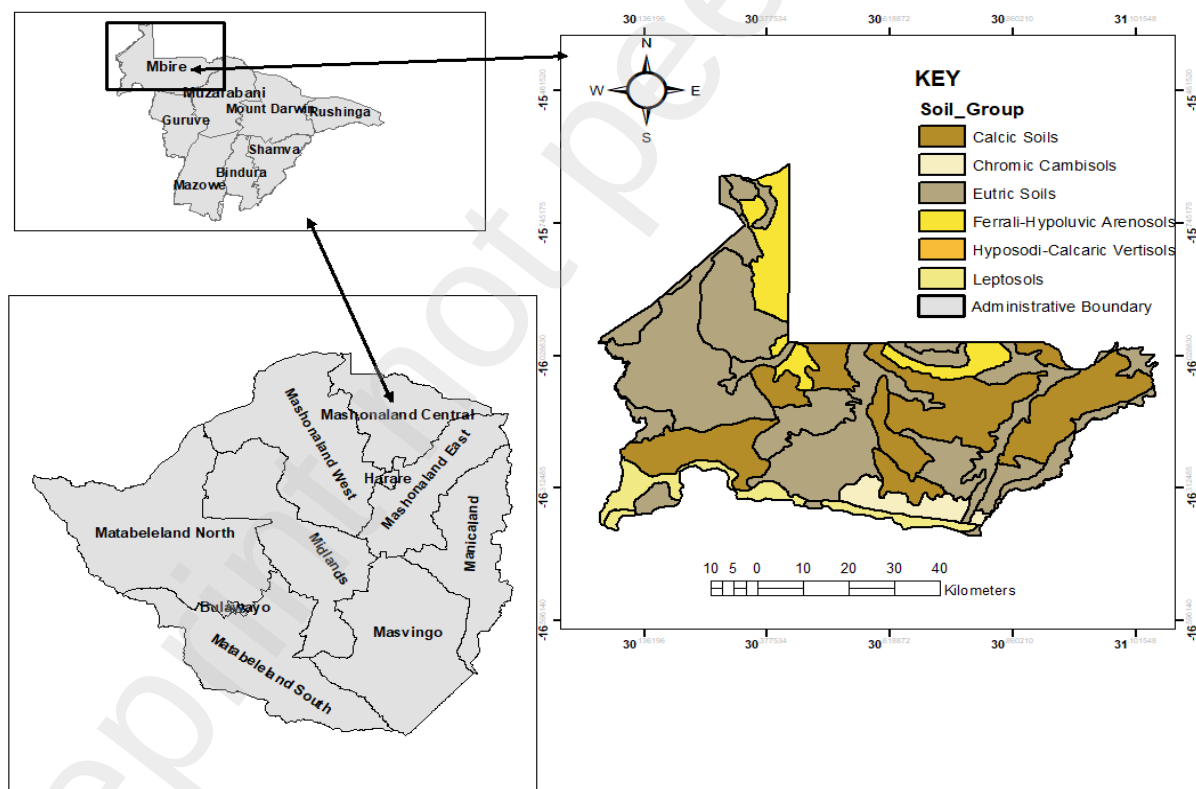


Figure 1: Map of Zimbabwe showing the Geographical Position of Mbire District

Mbire District, located in Mashonaland Central Province in northern Zimbabwe, provides a compelling setting for examining the role of government safety nets in climate adaptation and women's empowerment within agricultural communities. The district borders Zambia to the north, with the Zambezi River forming a natural boundary to the northwest. Covering a substantial land area, Mbire features diverse landscapes that include river valleys, floodplains, and mountainous terrain, all contributing to its unique environmental and socio-economic dynamics (Zimbabwe National Statistics Agency, 2022).

The natural resources of Mbire District make it particularly valuable for environmental and agricultural research. The district's biodiversity and ecosystem services are central to both conservation and sustainable development initiatives. Below are some of the environmental aspects of focus:

Biodiversity Conservation: Mbire is home to rich biodiversity, encompassing riverine forests, wetlands, and savannah grasslands that sustain numerous plant and animal species. These ecosystems offer critical habitats, supporting species diversity and ecological functions that are vital for the district's ecological health and resilience. Conservation efforts in the area could provide insights into habitat restoration and ecosystem management practices aimed at preserving biodiversity in similar semi-arid regions (Muboko and Murwira, 2019).

Climate Change Adaptation: The region faces significant climatic challenges, including erratic rainfall, prolonged droughts, and flooding episodes. Mbire's vulnerability to these climate threats necessitates research on adaptive strategies to mitigate impacts on agriculture and livelihoods. Local climate adaptation measures such as water resource management, climate-smart agriculture, and drought-resistant crop varieties are essential for building community resilience and fostering sustainable agricultural practices (Chanza and Gundu-Jakarasi, 2021).

The rich environmental context of Mbire District makes it a suitable case for studying the intersection of climate adaptation and women's empowerment within agriculture. The district's complex socio-ecological dynamics underscore the importance of targeted interventions that can enhance resilience to climate variability while promoting sustainable livelihood practices among local communities.

2. Research Methods

This study employed mixed-methods research design, integrating both qualitative and quantitative approaches. This combination provided a comprehensive understanding of food security issues by merging statistical data with qualitative narratives, which highlighted participants' personal experiences and perspectives. Mixed methods were chosen to offer a holistic view of the complexities surrounding food insecurity and the role of government safety nets in Mbire District. The use of this design is supported by Burns and Grove (2003), who emphasize the value of mixed methods in capturing the nuances of complex social issues through complementary data sources.

2.1 Study Population

The study focused on women actively engaged in various agricultural activities in Mbire District, Zimbabwe, with a deliberate effort to include women from diverse age groups, educational backgrounds, and agricultural experience levels. This diversity was crucial to understanding the differential impacts of government safety nets on climate adaptation and women's empowerment. Key stakeholders, such as government officials, NGO representatives involved in climate adaptation and women's empowerment, community leaders, and agricultural extension officers, were also included. This inclusion provided insights into the effectiveness and challenges of implementing safety nets. Additionally, men involved in agriculture were included to capture broader community perspectives, as they may also be directly or indirectly impacted by government interventions. This approach enabled a comprehensive and inclusive analysis of the research questions, ensuring that findings reflect the experiences of a wide range of individuals affected by government safety nets in Mbire District (Yin, 2014).

2.2 Sampling Procedures

A combination of purposive, quota, and snowball sampling techniques was employed to target women whose primary source of income is crop production. Purposive sampling allowed for the selection of participants based on specific criteria, ensuring that the sample represented the target population, specifically female farmers dependent on crop farming for food security and income. Quota sampling was employed to capture population diversity, representing different age groups and farming experience levels. Snowball sampling was used to reach isolated or hard-to-access participants through referrals from initial respondents, a particularly useful approach when seeking participation from marginalized or dispersed community members (Patton, 2015). The study aimed at a sample size of 250 participants. The collected responses were organized into themes, and both qualitative and quantitative data were analyzed and presented using graphs, tables, and verbatim statements from participants, enriching the overall data presentation.

2.3 Data Management and Analysis Methods

Data was collected, cleaned, and prepared for analysis following established data management processes. Using Kobo Collect, the data was saved, downloaded, and subsequently imported into SPSS for further processing. Sekaran and Bougie (2016) describe data management as encompassing the processes of cleaning, transforming, and modeling data to generate valuable insights, support decision-making, and informing research conclusions. In this study, descriptive statistics, presented in tables and charts (including pie charts), provided an overview of the data. Additionally, inferential tests such as t-tests, ANOVA, and regression models were conducted to assess associations and make predictions regarding the impact of social safety nets on food security and women's empowerment outcomes.

3. Results and Discussion

3.1 Introduction

The study distributed 255 questionnaires, with 250 completed and returned, resulting in a response rate of 98%. According to Creswell (2018), a response rate above 70% is typically sufficient for reliable data analysis, making this study's 98% highly robust for statistical evaluation (Creswell, 2018). Table 1 provides an overview of the response rates.

Table 1: Response Rate of Participants

Category	Frequency	Percentage (%)
Administered	255	100.0
Returned	250	98

Source: Primary Data, 2024

3.2 Demographic Profile of Respondents

An accurate demographic profile is essential to contextualize the sample and ensure its alignment with the target population.

Gender

This study focuses exclusively on women due to their significant roles in agricultural production and household food security in Mbire District. Women in the region bear a disproportionate burden of climate impacts, as they are primary caregivers and central to local agricultural activities. They are also often at the forefront of climate adaptation initiatives, necessitating an examination of the gender-specific challenges they face, and the opportunities government safety nets provide for their empowerment. Concentrating on female respondents highlights the unique barriers and adaptive strategies they employ, while recognizing that men also face climate-related challenges. This targeted approach is intended to amplify the insights relevant to women's empowerment and resilience in agricultural settings amid climate risks (FAO, 2019; UN Women, 2021).

Age Distribution

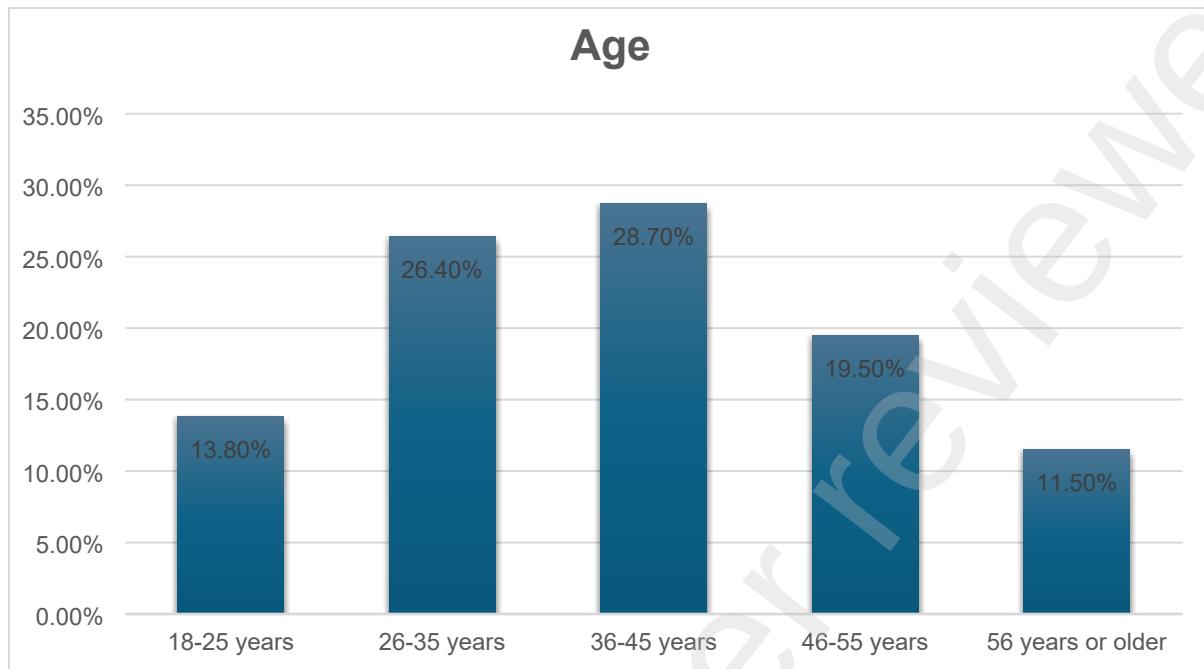


Figure 2: Respondents Age- Source: Field Data, 2024

The age distribution of respondents in Mbire District is diverse, spanning multiple age cohorts, which enriches the study with varied perspectives on climate adaptation and women's empowerment in agriculture. The most representative group, aged 36-45, comprises 28.7% of respondents, suggesting active involvement in household food security and climate adaptation strategies, often reflecting prime working-age individuals who bear significant domestic and economic responsibilities (Field Data, 2024). This aligns with studies indicating that this age group is crucial in implementing adaptive measures due to their experience and resource access (Nyanga, 2012).

The 26-35 and 46-55 age groups also show considerable participation, representing 26.4% and 19.5% of respondents, respectively. These age brackets may encounter specific challenges in balancing agricultural productivity with climate-related impacts, with the younger group potentially being more adaptive to new technologies (Mutekwa, 2009). Meanwhile, respondents aged 18-25, though fewer at 13.6%, contribute fresh perspectives and often more progressive views on adaptation practices, crucial for inter-generational knowledge transfer (FAO, 2017).

Participants aged 56 and above represent 11.5% of the sample. Despite their lower representation, this cohort brings essential insights rooted in long-term experiences with climate variability, often drawing on traditional practices that can enrich adaptation frameworks (Dube and Phiri, 2013). The diversity of age groups in the sample thus supports a comprehensive understanding of the community's adaptive capacity, reinforcing the need for age-sensitive climate adaptation policies and programs that leverage the strengths of different cohorts.

Level of Education

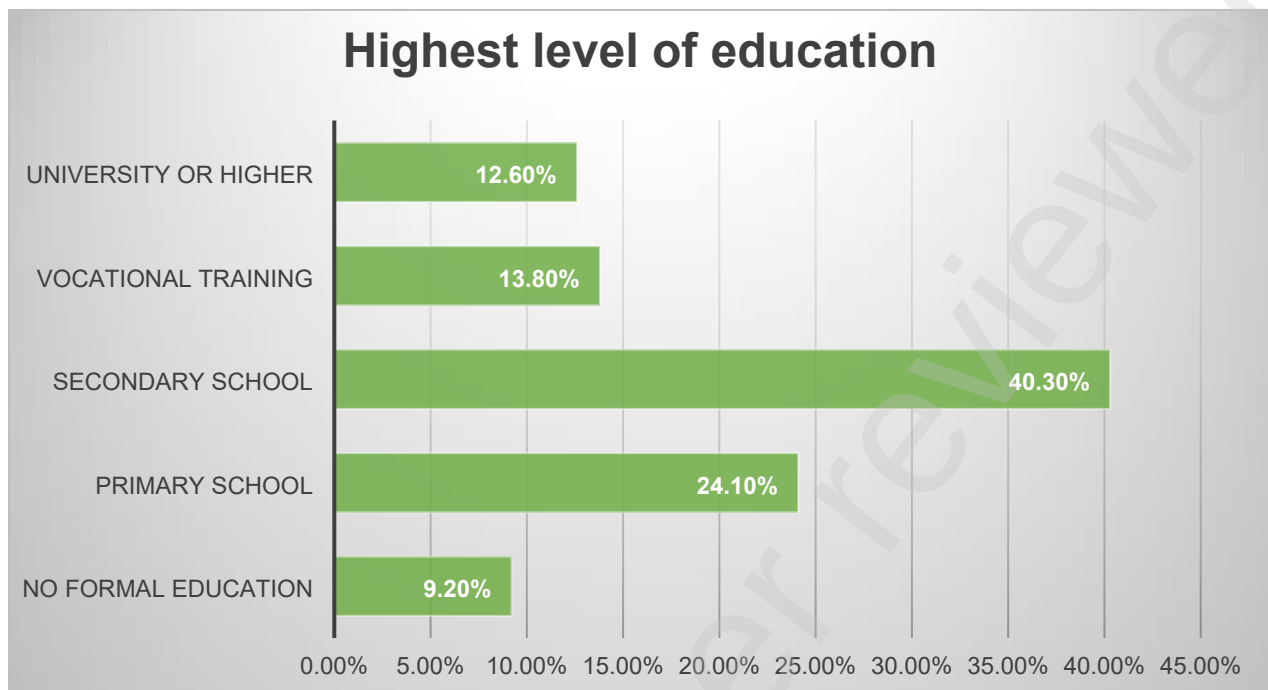


Figure 3: Educational qualifications- Source: Field Data, 2024

The educational qualifications of respondents, as presented in Figure 3 (Field data, 2024), indicate a diverse range within the sample. A significant portion (40.3%) had completed secondary education, suggesting a generally high level of formal education among respondents. This educational attainment may positively impact climate adaptation and agricultural productivity by increasing awareness and understanding of climate-resilient practices as also observed by Mutekwa, (2009).

However, the data also highlights an inclusion of marginalized groups: 9.2% had no formal education, while 24.1% had only completed primary school. This variation allows for a comprehensive assessment of adaptive capacities and household food security across different educational backgrounds, especially for individuals who may face constraints in accessing resources and information (Brown and Westaway, 2011).

Furthermore, the presence of respondents with vocational training (13.8%) and higher education (12.4%) contributes additional insights into the role of specialized skills and academic knowledge in shaping effective climate adaptation. These educational levels often enhance understanding of technical solutions and resource management, which are essential in adapting to climate impacts (Adger et al., 2009). Therefore, the sample's educational diversity is instrumental in evaluating the influence of education on climate adaptation strategies and women's empowerment in Mbire District.

3.3 Identification of Existing Social Safety Nets for Food Security Improvement

Table 2: Social Safety Nets Accessed by Households

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Food Aid Programmes	58	23.2	23.2	23.2
	Cash Transfer Programmes	32	12.8	12.8	36.0
	Public Works Programmes	2	.8	.8	36.8
	Subsidized Agricultural Inputs	95	38.0	38.0	74.8
	Non-Recipients	63	25.2	25.2	100.0
	Total	250	100.0	100.0	

Table 2 presents the distribution of social safety nets received by households in Mbire District. The most prevalent form of support is subsidized agricultural inputs, which accounts for 38.0% of respondents. This indicates a significant reliance on agricultural assistance as a key strategy for improving food security in the district. Food aid programmes follow closely, representing 23.2%, emphasizing the importance of addressing immediate food insecurity, although to a lesser extent than agricultural input support. Cash transfer programmes comprise 12.8%, reflecting moderate direct financial assistance. Public works programmes, at 0.8%, are the least utilized, suggesting limited engagement or effectiveness in this area. Notably, 25.2% of respondents report not receiving any form of social safety net, raising concerns about the accessibility and coverage of these programmes.

The dominance of agricultural subsidies and food aid suggests a focus on immediate food security and agricultural productivity. However, the exclusion of a significant proportion of the population underscores the need for broader and more inclusive safety net programs. Expanding coverage and enhancing the reach of these initiatives could help address gaps in support and improve resilience among vulnerable populations (Munyua, 2017; Chigonda and Chiremba, 2020). Furthermore, integrating more diverse and sustainable safety nets, such as cash transfers and public works, could strengthen the overall effectiveness of the programmes (Devereux and Sabates-Wheeler, 2004).

3.4 Impact of Social Safety Nets on Food Security in Mbire District

Table 3 the impact of these social safety nets on your household's food security

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Very High	81	32.4	43.3	43.3
	High	55	22.0	29.4	72.7
	Moderate	2	.8	1.1	73.8
	Very Low	49	19.6	26.2	100.0
	Total	187	74.8	100.0	
Missing	System	63	25.2		
Total		250	100.0		

Table 3 illustrates the perceived impact of social safety nets on household food security in Mbire District. A notable 43.3% of participants rated the impact as "Very High," reflecting a strong positive perception of the effectiveness of these safety nets in enhancing food security. Additionally, 32.4% of respondents assessed the impact as "High," and 22.0% rated it as "Moderate," indicating that most participants view these safety nets as beneficial to their food security. Conversely, only 0.8% rated the impact as "Low," and 19.6% perceived it as "Very Low."

These findings highlight a generally favorable perception of social safety nets, with the majority of participants acknowledging their critical role in improving household food security. The high percentage of "Very High" ratings reinforces the belief that these programs significantly contribute to alleviating food insecurity, aligning with studies that have demonstrated the positive outcomes of such safety nets in similar contexts (Cohen and Sebstad, 2005; Devereux, 2006).

3.5 Climate Change Adaptation Strategies

The analysis of climate change adaptation strategies in Mbire District, as shown in Table 4, reveals varying levels of agreement among respondents on different approaches.

3.5.1 Planting Drought-Resistant Crops

The most widely supported strategy was planting drought-resistant crops, with 71.2% of respondents agreeing or strongly agreeing. This indicates a clear recognition of the importance of crop varieties that can withstand water scarcity, a common challenge due to recurrent droughts. Drought-resistant crops are crucial for maintaining agricultural productivity under adverse climatic conditions, reinforcing the community's adaptive capacity (FAO, 2015).

3.5.2 Diversifying Crops and Livestock

Diversification was the second most supported strategy, with 67.8% of respondents in agreement. This reflects the community's understanding that diversifying both crops and livestock can

mitigate the risks posed by climate-related shocks, such as crop failure or livestock disease outbreaks (Sullivan, 2011). Diversification offers multiple income streams and enhances food security, particularly in unpredictable climates.

3.5.3 Implementing Water Conservation Practices

69.0% of respondents supported water conservation measures, recognizing their significance in ensuring sustainable water availability. This is especially critical during droughts or periods of water scarcity. Techniques such as rainwater harvesting and efficient irrigation practices are essential for maintaining water availability, ensuring that agriculture remains viable during dry spells (Krause et al., 2018).

3.5.4 Using Improved Irrigation Techniques

Approximately 66.7% of respondents agreed on the importance of adopting improved irrigation techniques. This highlights the community's awareness of the role efficient water management plays in enhancing crop productivity, particularly in areas facing water scarcity or erratic rainfall patterns (Molefe et al., 2017). Optimized irrigation is a key adaptation strategy to ensure agricultural resilience in arid regions.

3.5.5 Implementing Soil Conservation Measures

Soil conservation received support from 59.8% of respondents, indicating recognition of its importance in preserving soil health and fertility. Practices such as erosion control, organic farming, and proper land management help protect vital soil resources, which are foundational for long-term agricultural productivity and climate resilience (Pretty, 2011).

3.5.6 Agroforestry Practices

Agroforestry, though beneficial for soil conservation, microclimate regulation, and income diversification, was the least adopted strategy, with only 44.8% of respondents in agreement. This suggests that while its benefits are recognized, agroforestry may require further promotion and capacity-building initiatives to increase its adoption (Kandji et al., 2006). There is an opportunity to raise awareness about its potential advantages and integrate it into broader agricultural adaptation programmes.

Table 4: Climate Change Adaptation Strategies (n=250) Source: Field data, 2024

Adaptation Strategy	Strongly Disagree (%)	Disagree (%)	Neutral (%)	Agree (%)	Strongly Agree (%)
Planting drought-resistant crops	5.7	9.2	13.8	40.2	31.0
Implementing water conservation practices	3.4	10.3	17.2	43.7	25.3
Using improved irrigation techniques	8.0	16.1	21.8	31.0	23.0
Practicing agroforestry	11.5	17.2	26.4	25.3	19.5
Implementing soil conservation measures	6.9	12.6	20.7	36.8	23.0
Diversifying crops and livestock	4.6	9.2	18.4	39.1	28.7

Overall, the findings highlight a community well-engaged in climate change adaptation strategies, with particular emphasis on drought-resistant crops and diversification. However, less adoption of agroforestry indicates a need for further awareness and targeted interventions. These results can inform local policymakers and development practitioners of the most effective strategies to bolster climate resilience, providing a foundation for more tailored and inclusive interventions.

3.6 Alternative Strategies for Improving Food Security in Mbire District

Table 5 Strategies perceived to be more effective

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Improved agricultural training and support	72	28.8	28.8	28.8
	Enhanced market access for farmers	46	18.4	18.4	47.2
	Diversification of income sources	95	38.0	38.0	85.2
	Community-based food security programs	37	14.8	14.8	100.0
	Total	250	100.0	100.0	

Table 5 highlights the key strategies perceived to enhance food security and support climate adaptation and women's empowerment in agriculture within Mbire District. The survey results show that diversification of income sources was considered the most effective strategy (38.0%), followed by improved agricultural training and support (28.8%). Enhanced market access for farmers (18.4%) and community-based food security programmes (14.8%) were also identified as valuable approaches.

The prominence of income diversification aligns with findings in similar rural contexts, where diversified livelihoods have proven to mitigate climate-related risks by spreading economic dependency across multiple sources, thus increasing resilience (Agwu and Okhimamhe, 2009). Improved agricultural training and support further enhance adaptive capacities, particularly for women farmers who often lack access to technical knowledge and resources (Farnworth et al., 2015). Enhanced market access can empower women by enabling them to participate in value chains, increasing both income and decision-making power in household economies (Meinzen-Dick et al., 2011). Community-based food security programs, though rated lower, are essential for strengthening social capital and collective resilience, as also noted by Davies et al. (2008) in their study on community-driven adaptation strategies.

These findings underscore the importance of integrated and community-centered approaches in climate adaptation strategies, which not only improve food security but also enhance women's empowerment in agriculture.

3.7 Effectiveness of Adaptation Strategies

The analysis reveals mixed perceptions among Mbire District respondents regarding the effectiveness of adaptation strategies (see Table 6). Approximately 48.2% of participants agreed or strongly agreed that their adopted strategies reduced vulnerability to droughts, while 28.8% disagreed. This finding aligns with studies indicating that climate adaptation practices can enhance resilience when well-supported by local contexts and resources (Mubaya and Mafongoya, 2017).

The following analysis (Table 6) regarding the effectiveness of adaptation strategies, in Mbire obtains:

Table 6: Effectiveness of Adaptation Strategies (n=250)

Effectiveness Indicator	Strongly Disagree (%)	Disagree (%)	Neutral (%)	Agree (%)	Strongly Agree (%)
The strategies we have adopted have made us less vulnerable to droughts	11.5%	17.3%	23.0%	28.7%	19.5%
We have experienced improvements in crop yields since implementing climate change strategies	13.8%	20.7%	25.3%	23.0%	17.2%
Our household feels more prepared to handle extreme weather events	17.2%	23.0%	28.7%	17.2%	13.8%
The climate change strategies we use have reduced our reliance on food aid	20.7%	25.3%	23.0%	17.2%	13.8%
Overall, the strategies we have adopted have improved our household food security	23.0%	28.7%	23.0%	11.5%	13.8%

Source: Field data, 2024

3.7.1 Reduction in Vulnerability to Droughts

Nearly half of the respondents (48.2%) felt that adaptation measures helped reduce drought vulnerability, a crucial outcome in drought-prone Mbire. Adaptation efforts appear effective in moderating climate impacts, as similar findings in other African contexts suggest that increased drought resilience often leads to enhanced food security and agricultural productivity (UNDP, 2020). The observed reduction in drought vulnerability implies a potential decrease in the

community's dependency on external food aid, corroborating the argument that effective adaptation measures are integral for sustaining rural livelihoods (Nyanga et al., 2016).

3.7.2 Enhancements in Crop Yields

A notable 43.7% of respondents reported improvements in crop yields since implementing climate adaptation strategies, suggesting a significant positive perception of these measures. Enhanced crop yields are vital for food security and income generation, indicating that the strategies employed in Mbire District have effectively contributed to agricultural productivity, thereby strengthening livelihoods. Similar findings have highlighted how climate-smart agricultural practices can lead to sustainable yield increases and resilience to climate stressors (FAO, 2021).

3.7.3 Preparedness for Extreme Weather Events

About 41.4% of respondents felt better prepared to manage extreme weather events due to the adaptation strategies adopted. This increased sense of preparedness is essential for minimizing the impacts of hazards like floods and storms. The respondents' positive perceptions suggest that these interventions have bolstered community resilience and adaptive capacity, critical elements in climate adaptation efforts (IPCC, 2022). Effective preparation and risk reduction strategies, such as those seen in Mbire, contribute directly to the resilience of vulnerable populations facing recurring climate threats (Nguyen et al., 2020).

3.7.4 Reduced Dependence on Food Aid

While views were mixed, 45.9% of respondents noted a decreased reliance on food aid due to these adaptation strategies. This reduction underscores the potential for adaptive interventions to enhance food security and foster self-sufficiency, diminishing the need for external aid. Reduced dependency on food aid not only strengthens resilience but also empowers communities to better withstand climate-related challenges (World Bank, 2022).

3.7.5 Improvement in Household Food Security

With 51.7% of respondents acknowledging improvements in household food security, the majority perceive these adaptation strategies as effective in securing food availability, accessibility, and utilization. Household food security is integral to climate resilience, and these findings indicate that adaptation measures have directly benefited the community's stability and well-being. Similar studies affirm that improved food security reduces vulnerability to climate impacts, strengthening overall resilience (UNDP, 2021).

The responses highlight the effectiveness of climate adaptation strategies in Mbire District, demonstrating improvements in crop yields, preparedness for extreme weather events, reduced reliance on food aid, and enhanced household food security. The generally favorable perceptions among respondents point to the potential for scaling these strategies to similar settings, emphasizing the value of continued investment in climate resilience initiatives. Such initiatives are

instrumental in fostering sustainable development and empowering communities in the face of escalating climate risks.

3.8 Factors Influencing Adoption and Intensity of Adaptation Measures

This study examines key factors influencing the adoption and intensity of climate change adaptation measures among women in agriculture in Mbire District, Zimbabwe (see Table 7).

3.8.1 Access to Financial Resources

With a mean score of 4.2 (SD = 0.7), access to financial resources emerged as a critical determinant of adaptation capacity, with low variability in responses. This consensus underscores that financial limitations hinder farmers—particularly women—from implementing essential strategies like acquiring drought-resistant seeds, irrigation systems, and advanced technologies, ultimately constraining both climate resilience and economic empowerment. Government safety nets and microfinance initiatives could alleviate these barriers, fostering greater adaptation and empowerment in the agricultural sector (Chikozho, 2021; Makate et al., 2019).

3.8.2 Availability of Technical Support and Information

Respondents rated technical support and information moderately high (mean = 4.0, SD = 0.8), indicating it is widely recognized but inconsistently accessible. This finding suggests that, while some farmers benefit from extension services and training, others lack access to the technical knowledge required for effective climate-smart practices. Expanding these services and tailoring them to women farmers' needs could amplify adaptation success (Speranza, 2018).

3.8.3 Knowledge and Awareness of Climate Change Impacts

High levels of awareness (mean = 4.3, SD = 0.6) highlight the importance of knowledge in fostering proactive adaptation strategies. The low variance suggests a broad understanding of climate impacts, which is essential for motivating community-wide action. Strengthening local knowledge and awareness programs can further encourage adaptation adoption and resilience building (Nyanga, 2017; IPCC, 2022).

3.8.4 Access to Agricultural Inputs

With a mean score of 3.8 (SD = 0.9), access to inputs such as seeds and fertilizers is acknowledged as significant but inconsistently available. The variability in responses reflects disparities in access, likely due to logistical and economic constraints. Addressing these gaps through improved input distribution networks could enhance the uptake of climate-resilient practices (FAO, 2021).

3.8.5 Social and Cultural Norms

Social and cultural norms scored a mean of 3.5 (SD = 1.0), indicating a recognized but varied influence on adaptation behaviors. This variation may stem from gender roles, traditional farming practices, and community dynamics. Tailoring interventions to respect local norms and promote

inclusive participation could increase adoption rates and empower women in climate adaptation (Douxchamps et al., 2019).

Table 7: Factors Influencing Adoption and Intensity of Adaptation Measures			
Factor	N	Mean	Std. Deviation
Access to financial resources	250	4.2	0.755
The availability of technical support and information	250	4.0	0.863
Knowledge and awareness of climate change impacts	250	4.3	0.647
Access to agricultural inputs	250	3.8	0.971
Social and cultural norms	250	3.5	1.079
Valid N (listwise)	250		

Source: Field Data, 2024 (SPSS Version 20)

This analysis identifies key barriers and facilitators to climate adaptation in Mbire District, with financial resources, knowledge, and technical support as top priorities. Addressing these factors through government safety nets, tailored training, and access to agricultural inputs could significantly strengthen adaptation efforts and advance women's empowerment in agriculture. These findings underscore the need for targeted policies that support financial inclusion, knowledge dissemination, and the removal of social barriers, ensuring a sustainable and resilient agricultural sector in rural Zimbabwe.

3.9 Variations in Effectiveness of Climate Adaptation Measures Based on Socio-Economic Factors

The ANOVA results (Table 8) illustrate significant differences in the perceived effectiveness of climate change adaptation measures across socio-economic groups in Mbire District. The F-statistic (135.38) is highly significant ($p < 0.001$), confirming that socio-economic factors—

income levels, access to resources, and education levels—substantially influence adaptation outcomes.

Table 8: One Way ANOVA						
SUMMARY						
Groups	<i>Count</i>	<i>Sum</i>	<i>Average</i>	<i>Variance</i>		
Income levels	83	3170	38.13793103	39.76600985		
Access to resources	83	5046	60.79310345	24.0270936		
Education levels	82	6422	78.4137931	21.46551724		
ANOVA						
Source of Variation	<i>SS</i>	<i>Df</i>	<i>MS</i>	<i>F</i>	<i>P-value</i>	<i>F crit</i>
Between Groups	23643.6092	2	11821.8046	135.38	1.43E-24	3.105157
Within Groups	6983.94	80	87.298			
Total	30627.5632	82				

The between-groups sum of squares ($SS = 23643.61$) greatly exceeds the within-groups sum of squares ($SS = 6983.94$), indicating that socio-economic disparities contribute significantly to the variance in adaptation measure effectiveness.

Socio-Economic Influence on Adaptation Perceptions

Income Levels: Mean effectiveness rating of 38.14

Access to Resources: Mean effectiveness rating of 60.79

Education Levels: Mean effectiveness rating of 78.41

These findings suggest that individuals with higher income, better access to resources, and higher education perceive adaptation measures as notably more effective. This aligns with research highlighting that socio-economic empowerment enhances adaptive capacity, as observed in similar studies (e.g., Vincent et al., 2020; Nelson et al., 2018). Addressing resource constraints, enhancing education, and improving income opportunities are therefore critical for maximizing the effectiveness of climate adaptation strategies and fostering women's empowerment in agricultural resilience efforts.

3.10 The Role of Government-Initiated Social Safety Nets in Enhancing Food Security in Mbire District

Table 9 Relationship between social safety nets and food security

Correlations				
			B5government_initiated_social_safety_nets	FCS_Binary_cat
Spearman's rho	B5government_initiated_social_safety_nets	Correlation Coefficient	1.000	.874
		Sig. (2-tailed)	.	.000
		N	250	250
	FCS_Binary_cat	Correlation Coefficient	.874	1.000
		Sig. (2-tailed)	.000	.
		N	250	250
. Correlation is significant at the 0.01 level (2-tailed).				

The analysis reveals a significant positive correlation (Spearman's $\rho = 0.874$, $p < 0.01$) between government-initiated social safety nets and food security in Mbire District (Table 9). This high correlation coefficient implies that as more households participate in these safety nets, food security outcomes improve markedly. This relationship's statistical significance at the 0.01 level underscores the robustness of the findings, affirming that the observed correlation is highly unlikely to result from random variation.

These findings align with similar studies emphasizing the effectiveness of social safety nets in supporting vulnerable populations. For example, Devereux (2001) and FAO (2016) highlight that social safety nets play a vital role in mitigating food insecurity during climatic shocks, particularly in rural, agriculture-dependent communities.

In the context of Mbire, these safety nets have likely contributed to improved resilience against climate-induced food shortages by providing social protection that stabilizes food availability and access (Kabeer, 2012). Additionally, by enhancing food security, such programs support women's roles in agriculture, which can strengthen their economic empowerment and decision-making capacity (Holmes and Jones, 2013).

3.11 Mean Food Consumption Score (FCS) between Beneficiaries and Non-Beneficiaries

Table 10 Testing differences between the FCS means of beneficiaries and non-beneficiaries

Group Statistics											
		B5government_initiated_social_safety_nets		N		Mean		Std. Deviation		Std. Error	
FCS		Non-beneficiary		63		33.2381		6.22291		.78401	
		Beneficiary		187		64.3021		14.88724		1.08866	
Independent Samples Test											
				Levene's Test for Equality of Variances		t-test for Equality of Means					
				F		Sig.		t	df	Sig. (2-tailed)	Mean Difference
										Lower	Upper
FCS	Equal variances assumed	35.306	.000	-16.078	248	.000	-31.06404	1.93204	-34.86934	-27.25874	
	Equal variances not assumed			-23.155	237.397	.000	-31.06404	1.34159	-33.70699	-28.42110	

The analysis presented in Table 10 highlights a substantial and statistically significant difference in the mean Food Consumption Scores (FCS) between beneficiaries and non-beneficiaries of government-initiated social safety nets in Mbire District. Beneficiaries of these safety nets exhibited a notably higher mean FCS of 64.30 compared to non-beneficiaries, who averaged 33.24. This result is supported by the t-test, with a highly significant p-value ($p < 0.001$), indicating that the observed difference is not due to chance.

The higher FCS among beneficiaries suggests that government safety nets play a critical role in enhancing food security, which can directly support climate adaptation efforts in vulnerable agricultural communities (Adger, 2006). Food security, as a component of adaptive capacity, enables households to better withstand climate-related stresses (Thornton et al., 2014). Additionally, the increased FCS in beneficiary households may facilitate women's empowerment in agriculture by reducing the immediate burden of food insecurity, allowing women to participate more actively in productive roles (Meinzen-Dick et al., 2019). These findings underscore the dual

impact of social safety nets in promoting both food security and women's empowerment as pathways to climate resilience.

3.12 Model Fit and Significance of Government Safety Nets in Climate Adaptation and Women's Empowerment

Table 11 Model Fit Information

Model Fitting Information				
Model	Model Fitting Criteria	Likelihood Ratio Tests		
	-2 Log Likelihood	Chi-Square	Df	Sig.
Intercept Only	217.024			
Final	13.003	204.022	2	.000
Pseudo R-Square				
Cox and Snell			.558	
Nagelkerke			.745	
McFadden			.590	

The model fitting information (Table 11) strongly indicates that the final model provides a robust fit for the data. The likelihood ratio tests reveal significant model alignment, with a p-value less than 0.001, suggesting that the model captures critical relationships between government safety nets, climate adaptation, and women's empowerment in agriculture. The -2 Log Likelihood for the final model (13.003) is markedly lower than that of the intercept-only model (217.024), signifying a substantial improvement in fit when the predictors are included. The chi-square value of 204.022 with 2 degrees of freedom, along with a highly significant p-value, affirms the model's explanatory power.

Pseudo R-squared values provide further evidence of model strength, with Cox and Snell at 0.558, Nagelkerke at 0.745, and McFadden at 0.590, indicating a substantial proportion of variance in women's empowerment through agricultural adaptation explained by government safety nets. This outcome aligns with findings from past studies that highlight the critical role of safety nets in fostering climate resilience and advancing gender equity in resource-dependent sectors (FAO, 2021; Nyasimi et al., 2017).

3.13 Impact of Government Safety Nets on Household Food Security

Table 12 shows the relationship between government initiated social safety nets and food security in Mbire District.

Table 12 Multinomial test. Effect of FSC on Food security

Parameter Estimates									
FCS_cat ^a		B	Std . Error	Wald	df	Sig.	Exp(B)	95% Confidence Interval for Exp(B)	
								Lower Bound	Upper Bound
borderline food consumption	Intercept	-.916	.837	1.199	1	.273			
	[C7government_initiated_social_safety_nets=0]	3.076	.941	10.679	1	.001	21.667	3.425	137.076
	[C7government_initiated_social_safety_nets=1]	0 ^b	.	.	0	.	.	.	.
acceptable food consumption	Intercept	3.584	.453	62.473	1	.000			
	[C7government_initiated_social_safety_nets=0]	-3.766	.756	24.783	1	.000	.023	.005	.102
	[C7government_initiated_social_safety_nets=1]	0 ^b	.	.	0	.	.	.	.
a. The reference category is poor food consumption.									
b. This parameter is set to zero because it is redundant.									

The results reveal a statistically significant association between government-initiated social safety nets and household food security in Mbire District. As shown in Table 12, households with access to government safety nets experience enhanced food security, measured through food consumption scores (FCS).

Borderline Food Consumption: For households with borderline food consumption, the presence of government-initiated social safety nets (coded as 1) is associated with a positive coefficient (B = 3.076), indicating that these households have substantially higher odds (Exp(B) = 21.667, 95% CI: 3.425 - 137.076) of achieving acceptable food consumption compared to those without access to such safety nets. The Wald test confirms the statistical significance of this effect (p = 0.001), emphasizing the transformative role of safety nets in improving food security outcomes among vulnerable households.

Acceptable Food Consumption: Conversely, households with acceptable food consumption who receive government safety nets demonstrate a reduced risk of food security deterioration, as indicated by a negative coefficient ($B = -3.766$). This translates to a lower likelihood ($\text{Exp}(B) = 0.023$, 95% CI: 0.005 - 0.102) of regressing to poor food consumption compared to households without safety nets. The Wald test also validates this finding ($p = 0.000$), underlining the role of safety nets in preserving food security for households that have attained an acceptable food consumption level.

Overall, these findings support the hypothesis that government-initiated social safety nets significantly enhance food security for at-risk households. For those on the borderline, these interventions help advance food security status, while for those already secure, they offer protection against backsliding. This is consistent with previous studies indicating the positive impact of safety nets on food security resilience in other vulnerable regions (FAO, 2016; Chiweshe and Mutangi, 2020).

4. Discussions and Recommendations

The study evaluated the impact of government safety nets on climate change adaptation and women's empowerment in agriculture within Mbire District, Zimbabwe. Key findings and recommendations included:

4.1 Enhances Social Safety Nets for Food Security

Existing safety nets, including food aid, cash transfers, and agricultural input subsidies, significantly contribute to food security, with input subsidies being the most utilized (38%). This trend aligns with Jayne et al. (2010), who emphasize the widespread use of input subsidies in Africa to bolster food security. Evidence from Devereux and Sabates-Wheeler (2004) suggests that transformative social protection, encompassing subsidies and transfers, enhances food security, a finding reflected in Mbire's 12.8% household uptake of cash transfers (Handa et al., 2018). These programs underscore a multi-dimensional approach critical for food security (World Bank, 2020; UNICEF, 2018). To maximize impact in Mbire District, government and partners should work at expanding accessibility, that is, broadening access to cash transfers and subsidies to reach more households. In diversifying interventions, there is need to integrate multi-dimensional approaches combining cash inputs and training to strengthen food security resilience.

4.2 Promote Climate Change Adaptation Strategies

Households rely on strategies such as drought-resistant crops, diversification, and water conservation. Agro-forestry, though underutilized, presents untapped potential. There is need for capacity building, that is, training farmers on agroforestry and advanced irrigation techniques. Support infrastructure in terms of developing water conservation and irrigation systems to enhance long-term resilience should be prioritized.

4.3 Improve Effectiveness of Adaptation Strategies

Adaptation strategies reduce vulnerability, improve yields, and decrease reliance on food aid. To enhance effectiveness, sustained investment is necessary. Government should prioritize funding for proven strategies like crop diversification and drought-resistant technologies. Scaling efforts focusing on replicating successful interventions in similar drought prone areas should top the agenda of government and partners in the next five seasons.

4.4 Address Barriers to Adoption

Key adoption drivers include financial resources, technical support and access to information. To overcome these barriers there is need to continue with subsidized resources, that is, government and partners should provide affordable agricultural inputs and extension services. Cultural sensitivity will be promoted by designing interventions aligned with local norms and gender dynamics.

4.5 Mitigate Socio-Economic Disparities

Higher socio-economic status correlates with better adaptation outcomes, revealing disparities in resource access. To mitigate this challenge, there is need for targeted support. Focus in terms of targeting should be on marginalized groups to reduce inequalities. Education and training should be prioritized, Knowledge-sharing to empower resource-limited households ought to be promoted.

4.6 Leverage Social Safety Nets for Food Security

Social safety nets significantly enhance food security, as evidenced by higher food consumption scores among beneficiaries. To this end, the study recommends strengthening interventions by increasing funding and implementation efficiency to maximize reach and impact. Extension departments should regularly monitor and evaluate interventions and regularly assess intervention effectiveness to ensure continued improvement. Implementing these recommendations will ensure government can enhance the resilience of smallholder farmers in Mbire District, strengthen food security and empower women in agriculture, creating a model for climate adaptation in similar contexts.

5 Conclusion

This study reveals that government safety nets significantly contribute to climate change adaptation and enhance food security and women's empowerment in Mbire District. Effective adaptation strategies, particularly those enhancing crop resilience and diversification, are well-regarded within the community, promoting food security and climate resilience. The successful adoption of adaptation measures is shaped by financial resources, technical support, climate change knowledge, and socio-cultural factors, which collectively require targeted interventions for sustainable impact. Socio-economic disparities in effectiveness underscore the need for equitable resource distribution to empower vulnerable groups. Continued support, investment, and inclusive

approaches are essential for enhancing resilience and reducing climate change adaptation inequities.

Credit authorship contribution statement:

Lovemore Mafuva: Conceptualization, Methodology, Investigation, Writing-Original Draft

Everson Ndlovu: Validation, Methodology, Writing- Review and Editing, Supervision

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have influenced the work reported in this article.

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